

Understanding Analysis, 2nd Edition

Chapter 7 - Section 7.2 - The Riemann Integral

1. Let f be a bounded function on $[a, b]$, and let P be an arbitrary partition of $[a, b]$. First, explain why $U(f) \geq L(f, P)$. Now, prove Lemma 7.2.6: for any bounded function f on $[a, b]$, it is always the case that $U(f) \geq L(f)$.

Solution: We'll prove it by contradiction. Suppose that $U(f) < L(f, P)$. By definition of infimum, there must exist a partition P_2 such that $U(f, P_2) < L(f, P)$, for otherwise the infimum would've been at least $L(f, P)$.

But $U(f, P_2) < L(f, P)$ contradicts Lemma 7.2.4: $L(f, P_1) \leq U(f, P_2)$ for any partitions P_1, P_2 .

To use this fact to prove Lemma 7.2.6, note that $L(f)$ is the supremum of all lower sums while $U(f)$ is an upper bound. Since by definition, an upper bound is *at least* the supremum, it follows that $U(f) \geq L(f)$.

Alternatively, we can prove Lemma 7.2.6 directly by contradiction as well. Suppose $U(f) < L(f)$ and let m be the midpoint i.e. $m = \frac{L(f) + U(f)}{2}$. By definition of infimum, there exists a partition P_1 such that $U(f, P_1) < m$. By definition of supremum, there exists a partition P_2 such that $m < L(f, P_2)$. But $U(f, P_1) < m < L(f, P_2)$ contradicts Lemma 7.2.4.